

Sudden Cardiac Arrest During Sports Activity in Middle Age

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Background—Sports-associated sudden cardiac arrests (SCAs) occur mostly during middle age. We sought to determine the burden, characteristics, and outcomes of SCA during sports among middle-aged residents of a large US community.

Methods and Results—Patients with SCA who were 35 to 65 years of age were identified in a large, prospective, population-based study (2002–2013), with systematic and comprehensive assessment of their lifetime medical history. Of the 1247 SCA cases, 63 (5%) occurred during sports activities at a mean age of 51.1 ± 8.8 years, yielding an incidence of 21.7 (95% confidence interval, 8.1–35.4) per 1 million per year. The incidence varied significantly by sex, with a higher incidence among men (relative risk, 18.68; 95% confidence interval, 2.50–139.56) for sports SCAs compared with all other SCAs (relative risk 2.58; 95% confidence interval, 2.12–3.13). Sports SCA was also more likely to be a witnessed event (87% versus 53%; $P < 0.001$) with cardiopulmonary resuscitation (44% versus 25%; $P = 0.001$) and ventricular fibrillation (84% versus 51%; $P < 0.0001$). Survival to hospital discharge was higher for sports-associated SCA (23.2% versus 13.6%; $P = 0.04$). Sports SCA cases presented with known preexisting cardiac disease in 16% and ≥ 1 cardiovascular risk factors in 56%, and overall, 36% of cases had typical cardiovascular symptoms during the week preceding the SCA.

Conclusions—Sports-associated SCA in middle age represents a relatively small proportion of the overall SCA burden, reinforcing the idea of the high-benefit, low-risk nature of sports activity. Especially in light of current population aging trends, our findings emphasize that targeted education could maximize both safety and acceptance of sports activity in the older athlete. (*Circulation*. 2015;131:1384–1391. DOI: 10.1161/CIRCULATIONAHA.114.011988.)

Key Words: death, sudden ■ exercise ■ heart arrest ■ ventricular fibrillation

Sudden cardiac arrest (SCA) represents a major public health issue worldwide, accounting for almost half of cardiovascular mortality.^{1,2} When occurring in the setting of sports, SCA generates major societal attention and alarm because the collapse usually occurs in a public setting and in front of many witnesses, striking those considered to be the healthiest in the community.

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SCA among young competitive athletes has been extensively investigated in the United States,^{3,4} allowing significant improvements in the field, notably the initiation and optimization of preventive strategies.^{5–9} However, except for limited data on specific recreational sport activities,^{10,11} there are, to the best of our knowledge, no studies evaluating SCA during sports among middle-aged subjects in any US community. Recent European experience has emphasized that, in the community, the largest burden of sports-associated SCA results from events among middle-aged participants.^{12–15} A better understanding of the burden (absolute and relative to other non-sports-related [nonsports] SCAs) and characteristics

of SCA during sports in middle-aged population is likely to inform preventive strategies.

Available information has traditionally been restricted to resuscitation variables in the immediate setting of the event. However, for a study to be meaningful from a clinical perspective, a detailed evaluation of the lifetime medical history is needed, which is particularly challenging to obtain for SCA because the majority of SCA patients will inevitably die in the field. Consequently the information collected by emergency medical services (EMS) is often restricted to data on the resuscitation process. Details of the medical history of SCA patients (especially those who cannot be resuscitated) are thus usually sparse and very rarely considered systematically. In this regard, information concerning previous cardiovascular risk factors and the frequency and type of symptoms in the weeks preceding SCA during sports has not previously been characterized.

Using a large 12-year prospective study of SCA in a north-western US metropolitan region, with a systematic comprehensive evaluation of the lifetime clinical history, we assessed the characteristics and outcomes of SCA during sports in middle age compared with other SCAs (not occurring during sports activity).

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Methods

Setting, Definitions, and Survey Methods

The Oregon Sudden Unexpected Death Study (Oregon-SUDS) is an ongoing community-based prospective study of out-of-hospital SCA. Detailed methods have been published previously.^{16,17} Briefly, since February 1, 2002, cases of SCA in the Portland, OR, metropolitan area were identified by the use of multiple sources, including the EMS response, the medical examiner's office, and emergency departments of all of the 16 local hospitals. The county has a 2-tiered emergency medical response system with involvement of advanced cardiac life support-staffed fire engine companies and transporting ambulances.

A comprehensive evaluation is performed for each case of unexpected death, including analysis of the circumstances of arrest recorded by EMS personnel or the medical examiner, prearrest medical records, and available autopsy data. A process of in-house adjudication by 3 physicians is used to determine cases meeting the criteria for SCA. In the event of disagreement about a specific case, the determination is based on the majority opinion. In the present analysis, we included middle-aged men and women, defined as male and female subjects 35 to 65 years of age, ascertained from February 2002 to January 2013. This study was approved by the Institutional Review boards of Cedars-Sinai Medical Center, Oregon Health and Science University, and all participating hospitals.

SCA was defined as a sudden unexpected pulseless condition; if unwitnessed, subjects were included if observed alive and symptom free within 24 hours of their sudden arrest. During the adjudication process, patients with known noncardiac causes of arrest (eg, trauma, overdose, pulmonary embolism, cerebrovascular accident, and terminal illness such as cancer not in remission) were excluded.^{1,15} SCA during sports was defined as death occurring during sports activities or within 1 hour of cessation of sports activity. In most cases, the event occurred as an unexpected collapse during physical exertion. Of note, SCA occurring during physical activity but not in the setting of sports (typically SCA during sexual intercourse) was not considered sports-associated SCA. Previous medical history was obtained from a comprehensive assessment of inpatient and outpatient medical records or visits obtained from the hospital systems in the Portland metropolitan area.

Variables Assessed

Variables considered in the present analysis included demographics, circumstances of occurrence, and resuscitation data. In addition to detailed information obtained from EMS, index hospital stay, and medical examiner medical records, a working group of the Oregon-SUDS collected and assessed the lifetime medical history for both deceased patients and consent survivors, including assessment of symptoms before SCA.

Variables included the time and location of each event and clinical and demographic information (age, sex, personal history of known cardiovascular risk factors or heart disease, and any symptoms during the preceding week). Symptom assessment was derived from multiple sources, including family members at the scene of the SCA, witnesses, and survivors of SCA. In addition, available hospital and outpatient medical records for all subjects were systematically analyzed for information on symptoms. Symptoms were categorized as absent, present, or not evaluable on the basis of the extent of information available and the agreement of investigators. Coronary artery disease was defined as $\geq 50\%$ lumen stenosis on coronary angiogram before the SCA event or identified at autopsy. In addition, subjects with history of myocardial infarction or coronary revascularization (either surgical or percutaneous coronary intervention) were categorized as patients with coronary artery disease. History of heart failure was defined by a physician on the basis of medical records (in-hospital or outpatient visit) or low left ventricular ejection fraction from echocardiogram, angiogram, or radionuclide multigated acquisition before and unrelated to cardiac arrest. Other variables included data on the setting and nature of sports activity at the time of the event; the circumstances of collapse (during sports activity or immediately after), location type (sports facility or other area), and presence of

witnesses; detailed information on resuscitation, including response time (time from EMS call to EMS arrival on the scene), bystander cardiopulmonary resuscitation (CPR) initiation, the presenting cardiac rhythm (ventricular tachycardia/fibrillation, pulseless electric activity, or asystole), return of spontaneous circulation (defined as return of a palpable pulse in conjunction with a systolic blood pressure of >60 mmHg); and survival status to hospital discharge, available in the majority of admitted cases (96%). For survivors, neurological status at discharge was evaluated with the Cerebral Performance Categories score from medical records, and a comparative analysis was performed between sports-associated SCA and other SCA.¹⁴ The final variable was the autopsy report, if available.

Statistical Analysis

This report was prepared in compliance with the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) checklist for observational studies.¹⁸ The objective of the study was to provide comprehensive comparisons between sports SCA and nonsports SCA, including subject characteristics, survival to hospital discharge, and neurological outcomes. Case characteristics were reported as mean $\pm$ SD, proportion, median, and interquartile range, as appropriate. Comparisons between groups used the χ^2 test for categorical variables, the Fisher exact test when appropriate, and the Student *t* test or Mann-Whitney test for continuous variables.

For the calculation of incidence rates, the population was limited to Multnomah County, Oregon, the largest subset of the metropolitan area residents (population, 679 348, including 135 868 men and 133 598 women 35–65 years of age; 2003 US Census Bureau data for Multnomah County) and for the period of 2002 to 2005 (when all SCAs were recorded). All Multnomah County residents 35 to 65 years of age were included for the calculation of annual incidence rates in middle-aged men and women (calculated per year per million residents).

Predictors of survival to hospital discharge were studied among SCA cases with resuscitation attempted ($n=936$). Any variables in univariate analyses with a value of $P<0.15$ were included in the multivariate modeling. The association of SCA with favorable outcome (survival to hospital discharge) was thereafter adjusted for these identified confounding factors in a single multivariable logistic regression analysis, allowing estimation of odds ratios (ORs) and their 95% confidence intervals (CIs). As a sensitivity analysis, to better appreciate the likelihood for survival when SCA occurs in the setting of sports activities, we performed additional analysis using propensity score matching.¹⁹ We calculated each subject's propensity score as the predicted probability of having SCA during sports given the subject's measured covariates (age, sex, history of coronary artery disease, presence of witness, bystander CPR, ventricular fibrillation, delay of intervention). We matched each case to the control subject (ie, a patient who experienced a nonsports SCA) with the closest propensity score, which resulted in a data set with 66 (33 and 33) SCA-matched cases. All data were analyzed with STATA software version 11.0 (StataCorp, College Station, TX).

Results

Patient Characteristics and Circumstances of SCA Occurrence

Between February 2002 and January 2013, there were a total of 63 subjects, middle-aged patients with SCA that occurred during sports activities (Table 1), mainly during jogging (27%), basketball (17%), and cycling (14%) activities. Other sports activities included gym activities (11%), golfing (8%), volleyball (3%), tennis (3%), soccer (3%), and others (14%). Sports-associated SCA occurred during the actual sports activity in 45 cases (76%) and within the following hour in 14 (24%). More than half of the cases (58%) occurred in sports facilities such as a gym or a stadium, whereas the remainder

Table 1. Patient Characteristics for SCA Associated With Sports Versus Not Associated With Sports

	Sports-Associated SCA (n=63)	Non-Sports-Associated SCA (n=1184)	P Value
Demographic data			
Age, y			
Mean±SD, y	51.1±9	53.1±8	0.08
35–50 y, n (%)	39 (62)	648 (55)	0.26
50–65 y, n (%)	24 (38)	536 (45)	
Timing, n (%)			
Weekend	28 (44)	314 (27)	0.002
Time of day			<0.001
Morning (6 AM–noon)	16 (25)	358 (30)	
Afternoon (noon–6 PM)	23 (37)	268 (23)	
Night (6 PM–6 AM)	7 (11)	234 (20)	
Unknown	17 (27)	324 (27)	
Season, n (%)			0.24*
Summer	19 (30)	258 (22)	
Autumn	14 (22)	288 (24)	
Winter	12 (19)	340 (29)	
Spring	18 (29)	298 (25)	
Cardiovascular risk factors, n (%)			0.38*
0	18 (43)	354 (33)	
1	9 (19)	159 (15)	
>1	18 (38)	509 (48)	
Type of cardiovascular risk factor			
Diabetes mellitus, n (%)	4 (8)	315 (28)	0.002
Dyslipidemia, n (%)	16 (33)	290 (26)	0.27
Systemic hypertension, n (%)	19 (40)	528 (47)	0.28
Body mass index, kg/m ²	29±6	30±8	0.21
Smoking, n (%)	10 (31)	451 (61)	0.001
Known heart disease, n (%)†	10 (16)	358 (30)	0.01*
Coronary artery disease	5 (8)	220 (19)	0.03
Atrial fibrillation	3 (5)	64 (5)	0.82
Heart failure	3 (5)	185 (16)	0.02
Pacemaker	3 (5)	56 (5)	0.95
Implantable cardioverter-defibrillator	0 (0)	2 (0)	1

Percentages were calculated on the basis of the total number of known events. SCA indicates sudden cardiac arrest.

*Significant.

†Among consented survivors or deceased patients with medical records available (n=583, including 45 sports-associated SCAs).

(42%) occurred on the field (outside of sports facilities). Among the 28 sports-associated SCAs that occurred outside of sports facilities, 18 occurred in public parks, jogging trails, or other delineated areas.

Of the overall number of middle-aged SCA cases in this region during the 10-year period (n=1247), sports-associated SCA in middle age represented 5% of SCA cases in this

age group, yielding an overall incidence of 21.7 (95% CI, 8.1–35.4) compared with 555.0 (95% CI, 541.4–568.7) per 1 million per year for nonsports SCA. The incidence varied significantly according to sex, with a much higher incidence among men (versus women) for sports-associated SCA (relative risk, 18.68; 95% CI, 2.50–139.56), as opposed to all other SCA for which the differences were smaller (relative risk, 2.58; 95% CI, 2.12–3.13). Extrapolating to the overall population of the United States, we estimated the total number of sports-associated SCA among the 35- to 65-year-old age group to be 2269 (95% CI, 772–3773) events among men and 136 (95% CI, 0–1846) events among women per year.

The mean age of patients developing SCA during sports was 51.1±8.8 years, without a significant difference between men and women (51.1±9.1 versus 50.3±4.6 years; $P=0.86$). Although there was an apparent similar overall prevalence of cardiovascular risk factors (≥1 cardiovascular risk factors, 56.3% versus 62.7%; $P=0.40$), sports SCA patients were less likely to present with smoking (31.3% versus 60.8%; $P=0.001$) and diabetes mellitus (8.3% versus 28.4%; $P=0.002$). Sports SCA patients were also less likely to have known heart disease compared with individuals experiencing nonsports SCA (15.9% versus 30.2%; $P=0.013$; Table 1). Among the 44 middle-aged participants with possible assessment of symptoms during the week before the event, 14 (33%) patients presented with symptoms: typical chest pain in 9 patients, dyspnea in 4 patients, and typical influenza-like symptoms in 1 patient. Although there were significant sex differences in the incidence of sports-associated SCA with a higher burden in men, no significant differences were identified between men and women for clinical characteristics, circumstances, and outcomes.

Bystander CPR, Advanced Life Support, and Predictors of Survival

Descriptive data on resuscitation and survival after SCA, based on the association with sports, are presented in Table 2. Compared with other SCAs, SCAs during sports were characterized by a greater proportion of witnessed events (87% versus 52%; $P<0.001$) and bystander CPR (44% versus 25%; $P=0.001$), as well as a greater likelihood of presenting with a shockable rhythm (84% versus 51%; $P<0.001$). The mean response time (6.8±4 minutes) was very similar between both groups ($P=0.99$), with the proportion of patients receiving intervention within 8 minutes being 77% and 79% in sports SCA and nonsports SCA, respectively ($P=0.44$). Survival to hospital discharge was higher for sports SCA (23.2%; 95% CI, 11.8–34.6) compared with nonsports SCA (13.6%; 95% CI, 11.6–15.5; $P=0.04$; Figure 1A).

Among overall SCAs with resuscitation attempted, variables independently (multivariate analysis) associated with survival included public location (OR, 1.55; 95% CI, 1.10–2.17; $P=0.011$), presence of witness (OR, 3.22; 95% CI, 1.80–5.77; $P<0.0001$), and initial shockable rhythm (OR, 4.54; 95% CI, 2.94–7.14; $P<0.0001$). SCA in the setting of sports activity was not associated with higher survival after adjustment for resuscitation variables (OR, 0.82; 95% CI, 0.36–1.82; $P=0.62$; Figure 1B). From the additional sensitivity analysis, the propensity score-matched OR for survival was not significantly

associated with sports SCA (OR, 0.84; 95% CI, 0.19–3.80; $P=0.82$). In addition, no significant association was observed for age (≤ 50 versus >50 years; OR, 1.13; 95% CI, 0.74–1.74; $P=0.55$) and delay in EMS response (<8 versus ≥ 8 minutes; OR, 0.96; 95% CI, 0.69–1.34; $P=0.83$).

Cause of SCA could be identified in 829 patients (66%), with no significant difference between sports SCA and nonsports SCA (68% versus 66%, respectively; $P=0.76$). The overall patterns of causes of SCA among sports and nonsports SCA are relatively similar (Figure 2), with a predominance of coronary artery disease. Among the 829 cases, 36 of 43 sports SCAs (84%) were associated with coronary artery disease compared with 717 of 786 for nonsports SCAs (91%). The phenotype of coronary artery disease among SCA cases, based on the setting of occurrence (sports versus nonsports), is reported in Table 3 for 181 cases (24%) overall with typical features of acute coronary syndrome (either on cardiac pathology or coronary angiogram). Among those subjects with known coronary artery disease (517 subjects), 140 (27%) had undergone a revascularization procedure (percutaneous coronary intervention in 46%) before developing SCA. Other causes of SCA included dilated cardiomyopathy (30), hypertrophic cardiomyopathy (30), congenital heart disease (6), mitral valve prolapse (3), myocarditis (3), and arrhythmogenic right ventricular dysplasia (2; Figure 2).

Discussion

To the best of our knowledge, this study represents the first comprehensive assessment of SCA associated with sports activity among middle-aged participants in a US community. First, our findings emphasize that sports-associated SCA

Table 2. Setting and Outcomes of SCA Associated With Sports Versus Not Associated With Sports

	Sports-Associated SCA (n=63)	Non-Sports-Associated SCA (n=1184)	P Value
Public occurrence, n (%)	55 (90)	242 (22)	<0.001
Sudden death witnessed, n (%)	55 (87)	614 (52)	<0.001
Bystander CPR, n (%)	28 (44)	300 (25)	0.001
First rhythm recorded on preadmission ECG, n (%)			<0.001
Ventricular fibrillation	42 (84)	437 (51)	
Asystole	4 (8)	180 (21)	
Pulseless electric activity	4 (8)	228 (27)	
Undetermined	0 (0)	9 (1)	
Call to EMS arrival			
Mean $\pm$ SD, min	6.79 $\pm$ 4	6.80 $\pm$ 4	0.99
≤ 8 min, n (%)	34 (77)	631 (79)	0.44
ROSC, n (%)	24 (39)	241 (29)	0.11
Survival to hospital discharge, % (95% CI)	23.2 (11.8–34.6)	13.6 (11.6–15.5)	0.04

Percentages were calculated on the basis of the total number of known events. CI indicates confidence interval; CPR, cardiopulmonary resuscitation; EMS, emergency medical services; ROSC, return of spontaneous circulation; and SCA, sudden cardiac arrest.

eventually represented a relatively small proportion of overall SCA in this population, a fact that is likely to encourage sports activities among middle-aged athletes after appropriate attention to any existing musculoskeletal limitations. Second, the comprehensive assessment of the lifetime medical history revealed that in almost two third of cases, patients present with either a previously documented significant cardiovascular disease or symptoms before the event. This latter finding suggests that the use of educational and general awareness approaches may further decrease the burden of sports SCA.

Due to a disproportionate emotional and lay press response to sudden death during sports, quantification of the burden requires an objective assessment. The design of our study, based on multiple sources of ascertainment, was especially useful for ensuring that all cases of interest were captured. The incidence of sports SCA estimated in this study is highly similar to recently reported experiences among men in Europe (from 33–58 per 1 million per year).^{12,13} However, by evaluating sports-associated SCA and overall SCAs in the same time period and geographical area, we were able to place the burden of sports SCA in the context of overall SCA burden and report that it eventually represented a small proportion of cases (1 of 20 SCAs). This finding suggests that in the context of SCA, the overall risk-to-benefit ratio is in favor of sports activity.

In our population, the incidence ratio between men and women was 7-fold higher in sports SCA compared with nonsports SCA. Whether this reflects an intrinsic difference in pathophysiology between sexes is still speculative, but some factors merit consideration. First, men may have higher sports participation rates than women, which may contribute to the sex differences in SCA rates. Consistent with this observation, the incidence of SCA in women was transiently elevated during moderate to vigorous exertion but with a relative risk of only 2.38 (95% CI, 1.23–4.60) while being almost 20-fold higher (relative risk, 44.9; 95% CI, 26.7–75.4) during vigorous activity among men enrolled in the Physicians' Health Study.^{20,21} Similar sex differences have also been reported in smaller retrospective studies in Finland and in the United States.^{22,23} Second, the duration of sports sessions and the vigor and the method of physical activity could differ between men and women. However, when inherent sex differences are considered, potential explanations could include a number of factors such as differences in vulnerable substrate (underlying structural or electric heart disorder), triggers, or autonomic modulators. The age-specific prevalence of coronary heart disease is known to be lower in middle-aged women (compared with men), and a particularly low prevalence has been observed among female victims of SCA, suggesting that women may have lower rates of sports SCA with myocardial ischemia as the primary cause.²⁴ Another interesting factor relates to coronary plaque rupture, which has been associated with a significant proportion of sports-related acute coronary syndrome and SCA.²⁵ Overall, women are significantly more likely to have plaque erosions (as opposed to plaque rupture) than men, potentially suggesting an inherently lower likelihood of plaque rupture in the setting of exercise in women. Sex-related differences in ventricular repolarization and vagal tone are also potential factors affecting sports SCA pathophysiology.²⁵

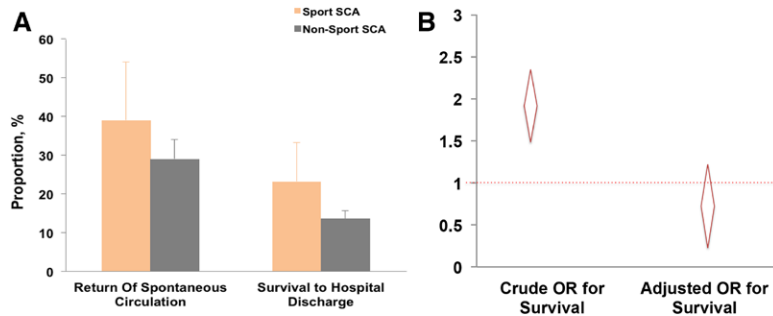


Figure 1. **A**, Proportion of patients with return of spontaneous circulation and survival to hospital discharge according to sudden cardiac arrest (SCA) occurring during sports or not during sports. **B**, Odds ratio (OR) for survival to hospital discharge of sports-associated SCA compared with non-sports-related SCA (crude and adjusted for age, sex, initial rhythm, response time, presence of witness, bystander cardiopulmonary resuscitation, and ventricular fibrillation). Error bars indicate upper 95% confidence interval.

Subjects who experience SCA during sports activity are generally perceived as previously healthy individuals with the event being considered truly “unexpected.” Our comprehensive assessment of the medical history in these patients demonstrates that in more than half, subjects had at least 1 cardiovascular risk factor. However, it is important to bear in mind that the long-term benefits of regular exercise on cardiovascular health are well established,^{21,26} and habitual physical activity has been demonstrated to reduce coronary heart disease events. Although existence of a “sports paradox”—vigorous activity can also acutely and transiently increase the risk of SCA and acute myocardial infarction, especially in those not habituated to exercise—has been recognized, overall, the long-term benefits outweigh the short-term risks related to physical activity.²¹ Thus, the findings from this study should in no way discourage patients with cardiovascular risk factors from engaging in regular, appropriate physical exercise within a framework of simple guiding rules from the treating physician. Whether a cautionary note needs to be raised with respect to sports in those with established coronary artery disease is a clinical question that needs further investigation. Physical activity counseling must be tailored to the needs and circumstances of the individual, thereby increasing the likelihood of success and decreasing the risk of cardiovascular

events during vigorous exercise.^{27–29} In this context, educating patients with recognized cardiovascular disease or risk factors on the avoidance of extreme exercise may minimize any potential harm²³ while maximizing the acceptance of sports activity as a tool to improve cardiovascular fitness.²¹

Of interest, the proportion of patients with cardiovascular risk factors may appear higher compared with the first published community-based report on SCA during sports in France (11%).¹² Although differences in the subjects’ characteristics may partly explain such differences, the lower rates in Europe may reflect the challenges in retrieving the detailed lifetime medical history in some studies. In addition, a significant proportion of the middle-aged subjects who eventually died during sports activity exhibited cardiac symptoms within a short period (1 week) preceding the SCA event. Taken together, these findings suggest that careful attention to preexisting heart disease and warning signs, along with public education and awareness in this regard, may have an important place in the overall strategy to reduce SCA associated with sports activity.

Overall, only one third of witnessed SCA cases received bystander CPR, indicating substantial room for improvement in public knowledge and attitudes toward basic life support.^{30–33} Bystander CPR is more likely to be initiated for SCA during sports, and this may have contributed to the higher likelihood of shockable rhythms at arrival of EMS, also extending the window of opportunity for successful defibrillation.^{34,35} The sports-associated SCA setting has appeal for the public and can serve as a springboard to efficiently educate the public on basic life support. Therefore, these findings also have potential implications for prehospital emergency care policy making to promote community education in basic life support.^{36,37} Although SCA during sports is associated with a higher survival,^{38,39} it seems that this is explained mainly by the concomitant association with factors that have a positive impact on the resuscitation process such as presentation with a shockable rhythm and the presence of witnesses during the SCA. Sports facilities have been identified as being particularly suitable for automated external defibrillator (AED) deployment⁴⁰ and where AED placement has largely been considered advisable.⁴¹ Our data further emphasize the need for systematic availability of AEDs in public sports facilities.⁴² However, because a good proportion of SCAs occurred outside sports facilities, there is still room for improvement in public AED deployment. One potential strategy under consideration is AED placement in strategic locations such as public parks and popular jogging trails. Additionally, public education to increase awareness also has an important role in improving bystander CPR.

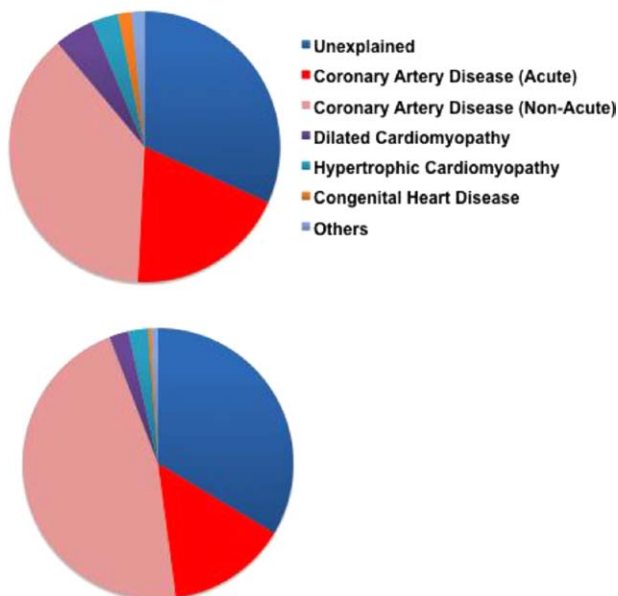


Figure 2. Distribution of cardiovascular abnormalities associated with sudden cardiac arrest (SCA) in the 1247 subjects according to sports SCA (top, n=63) and nonsports SCA (bottom; n=1184).

Table 3. Coronary Artery Disease Associated With SCA: Sports Versus Not Associated With Sports

	Sports-Associated SCA (n=63)	Non-Sports- Associated SCA (n=1184)	P Value
SCA with cause assessed, n (%)	43 (68)	786 (66)	0.89
Frequency of CAD, n (%)	36 (84)	717 (91)	0.10
Coronary presentation, n (%)			0.23
Acute MI	12 (33)	169 (24)	
History of CAD	3 (25)	69 (41)	
No history	9 (75)	100 (59)	
No acute MI,	24 (66)	548 (76)	
History of CAD	17 (71)	428 (78)	
No history	7 (29)	120 (22)	
History of CAD, n (%)	20 (56)	497 (69)	0.08
Previous revascularization	2 (10)	138 (28)	0.12
Surgical	1 (5)	74 (15)	
PCI	1 (5)	64 (13)	
History of angina	0 (0)	22 (4)	0.99
History of MI	12 (60)	246 (49)	0.37
Mean EF	48.1±18	47.1±16	0.84

Percentages were calculated on the basis of the total number of known events. CAD indicates coronary artery disease; EF, left ventricular ejection fraction; MI, myocardial infarction; PCI, percutaneous coronary intervention; and SCA, sudden cardiac arrest.

Although our data are, to the best of our knowledge, the first to address the issue related to burden and outcomes among the middle age from the general population in a large US community, with a particular efforts to provide data on past medical history, we acknowledge some limitations. First, because of the extremely low incidence of SCA among women, the statistical power to assess sex-related differences in sports SCA was limited. Second, we do not have access to information on whether individuals with sports-associated SCA underwent regular training before the SCA. Regular training has been shown to reduce the incidence of SCA during sports activity.²¹ Third, we are not able to provide details on the level of exercise at the time of the SCA for the entire period. Because there is no evidence to support a triggering effect of light exertion for SCA,²⁰ it may be possible that some of SCAs occurring during sports activities were associated by chance and not related to the activity itself, hence our use of the term sports-associated rather than sports-related SCA. For the SCA cases that occurred between 2002 and 2006, an estimated metabolic equivalent (MET) score, reflecting the intensity of exercise, was calculated.^{43,44} One MET is the amount of energy spent by a person sitting quietly. Physical activities were classified into 3 groups: light (<4 METs), moderate (4–6 METs), and heavy (>6 METs) sport activity. Overall, among the 30 SCAs during sports activities with estimated METs available between 2002 and 2006, light exertion was observed in 14% of SCA

cases, whereas moderate and vigorous were observed in 50% and 38%, respectively. Finally, specific coronary pathology examination could provide important information, especially details on potential sex-specific abnormalities, but was not routinely performed by the medical examiner in Portland.

Conclusions

The burden of sports-associated SCA among middle-aged participants is relatively low compared with the overall SCA burden in the community. Our findings of a high prevalence of established cardiovascular disease and symptoms that manifested in advance of SCA highlight a prevention gap that can potentially be closed. The significant prevalence of known cardiovascular disease and previous symptoms in affected participants offers an opportunity for targeted education to maximize both the safety and acceptance of sports activity in the middle-aged group.

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Disclosures

None.

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CLINICAL PERSPECTIVE

We performed a comprehensive study of sudden cardiac arrest during sports activity among all residents in a large US community of ≈ 1 million people. We report that, although the vast majority of sports-associated sudden cardiac arrest cases occur during middle age, they represent a relatively small proportion (5%) of overall sudden cardiac arrest cases. Sports-associated sudden cardiac arrest was 20-fold more common in men compared with women and associated with 2-fold higher survival rate compared with all other sudden cardiac arrests, likely because of the public nature of the setting. Sports-associated SCA cases also had a 2-fold lower rate of preexisting cardiac disease (16%), and overall, 36% had typical cardiovascular symptoms within the week preceding the SCA event. These findings reinforce the high-benefit, low-risk nature of sports activity. The discovery that many middle-aged athletes had known cardiovascular disease and experienced previous symptoms offers an opportunity for targeted education, with the goal of maximizing both safety and acceptance of sports activity in the older individual.